

QUESTION PAPER - SET 6

SECTION 1 - PERSONAL DATA

1. Name

2. Mobile Number

3. Class 10 Overall Percentage

4. Class 12 PCM Percentage

5. What is your percentage in Pre Sea Training till 4th Semester?

6. Tell us about your strengths.

7. Tell us about your weaknesses.

8. Have you participated in any extracurricular activities during your Pre Sea Training? If yes, share details.

9. Tell us about your family.

10. Where do you see yourself after 5 years?

SECTION 2 - COMPREHENSION PASSAGE

Archaeology as a profession faces two major problems.

First, it is the poorest of the poor. Only paltry sums are available for excavating and even less is available for publishing the results and preserving the sites once excavated. Yet archaeologists deal with priceless objects every day.

Second, there is the problem of illegal excavation, resulting in museum-quality pieces being sold to the highest bidder.

I would like to make an outrageous suggestion that would at one stroke provide funds for archaeology and reduce the amount of illegal digging. I would propose that scientific archaeological expeditions and governmental authorities sell excavated artifacts on the open market. Such sales would provide substantial funds for the excavation and preservation of archaeological sites and the publication of results. At the same time, they would break the illegal excavator's grip on the market, thereby decreasing the inducement to engage in illegal activities.

You might object that professionals excavate to acquire knowledge, not money. Moreover, ancient artifacts are part of our global cultural heritage, which should be available for all to appreciate, not sold to the highest bidder. I agree. Sell nothing that has unique artistic merit or scientific value. But, you might reply, everything that comes out of the ground has scientific value. Here we part company. Theoretically, you may be correct in claiming that every artifact has potential scientific value. Practically, you are wrong.

I refer to the thousands of pottery vessels and ancient lamps that are essentially duplicates of one another. In one small excavation in Cyprus, archaeologists recently uncovered 2,000 virtually indistinguishable small jugs in a single courtyard, even precious royal seal impressions known as melek handles have been found in abundance — more than 4,000 examples so far.

The basement of museums is simply not large enough to store the artifacts that are likely to be discovered in the future. There is not enough money even to catalogue the finds; as a result, they cannot be found again and become as inaccessible as if they had never been discovered. Indeed, with the help of a computer, sold artifacts could be more accessible than are the pieces stored in bulging museum basements. Prior to sale, each could be photographed and the list of the purchasers could be maintained on the computer. A purchaser could even be required to agree to return the piece if it should become needed for scientific purposes. It would be unrealistic to suggest that illegal digging would stop if artifacts were sold in the open market. But the demand for the clandestine product would be substantially reduced. Who would want an unmarked pot when another was available whose provenance was known, and that was dated stratigraphically by the professional archaeologist who excavated it?

1. The primary purpose of the passage is to propose

- A. an alternative to museum display of artifacts
- B. a way to curb illegal digging while benefiting the archaeological profession
- C. a way to distinguish artifacts with scientific value from those that have no such value
- D. the governmental regulation of archaeological sites
- E. a new system for cataloging duplicate artifacts

2. The author implies that all of the following statements about duplicate artifacts are true EXCEPT:

- A. A market for such artifacts already exists.
- B. Such artifacts seldom have scientific value.
- C. There is likely to be a continuing supply of such artifacts.
- D. Museums are well supplied with examples of such artifacts.
- E. Such artifacts frequently exceed in quality in comparison to those already cataloged in museum collections

3. Which of the following is mentioned in the passage as a disadvantage of storing artifacts in museum basements?

- A. Museum officials rarely allow scholars access to such artifacts.
- B. Space that could be better used for display is taken up for storage.
- C. Artifacts discovered in one excavation often become separated from each other.
- D. Such artifacts are often damaged by variations in temperature and humidity.
- E. Such artifacts' often remain uncatalogued and thus cannot be located once they are put in storage

4. The author's argument concerning the effect of the official sale of duplicate artifacts on illegal excavation is based on which of the following assumptions?

- A. Prospective purchasers would prefer to buy authenticated artifacts.
- B. The price of illegally excavated artifacts would rise.
- C. Computers could be used to trace sold artifacts.
- D. Illegal excavators would be forced to sell only duplicate artifacts.
- E. Money gained from selling authenticated artifacts could be used to investigate and prosecute illegal excavators

5. The author anticipates which of the following initial objections to the adoption of his proposal?

- A. Museum officials will become unwilling to store artifacts.
- B. An oversupply of salable artifacts will result and the demand for them will fall.
- C. Artifacts that would have been displayed in public places will be sold to private collectors.
- D. Illegal excavators will have an even larger supply of artifacts for resale.
- E. Counterfeiting of artifacts will become more commonplace

SECTION 3 - COMPREHENSION PASSAGE

The Sundarbans, located in the delta region of India and Bangladesh, is the largest mangrove forest in the world. It is home to diverse wildlife, including the famous Royal Bengal Tiger. The mangroves act as a natural barrier, protecting coastal areas from cyclones and storm surges. They also provide livelihoods for local communities through fishing, honey collection, and eco-tourism. However, climate change poses a serious threat to the Sundarbans. Rising sea levels, increasing salinity, and frequent storms are causing habitat loss.

Conservation projects are being implemented, focusing on planting more mangroves, reducing illegal logging, and promoting sustainable fishing practices. International cooperation is considered vital for the long-term preservation of this unique ecosystem.

1. Which of the following is NOT listed as a livelihood activity for local communities in the Sundarbans, according to the passage? (1 Point) Select the option that is not mentioned as a source of livelihood.

- A. Agriculture
- B. Honey collection
- C. Fishing
- D. Eco-tourism

2. Which of the following is a direct impact of climate change on the Sundarbans ecosystem? (1 Point) Select the option that best represents a climate change effect described in the passage.

- A. Increasing salinity of water
- B. Introduction of new wildlife species
- C. Growth in honey collection
- D. Expansion of fishing zones

3. What is one of the key protective roles played by the mangrove forests in the Sundarbans? (1 Point) Select the option that best describes how mangroves benefit the coastal environment.

- A. They increase the frequency of tidal waves.
- B. They are a major source of underground minerals.
- C. They supply drinking water to nearby villages.
- D. They act as a shield against cyclones and storm surges.

4. What is the primary objective of the conservation efforts described in the Sundarbans passage? (1 Point) Select the option that best reflects the main goal of these projects.

- A. To increase honey production for export
- B. To encourage local communities to move to urban areas
- C. To restore and safeguard the mangrove forests and their biodiversity
- D. To expand commercial fishing operations

5. Why does the passage emphasize the necessity of international cooperation for preserving the Sundarbans? (1 Point) Select the option that best explains the importance of cross-border collaboration.

- A. The Sundarbans ecosystem spans across India and Bangladesh, requiring joint efforts.
- B. Local communities are unwilling to support conservation projects.
- C. The region attracts international tourists who fund conservation.
- D. Climate change is only a concern for a few developed countries.

SECTION 4 - ENGLISH APTITUDE

1. Choose the correct transformation for the sentence: "The weather was too rough for the pilot to board."

- A. The weather was so rough that the pilot could board.
- B. The weather was rough enough for the pilot to board.
- C. The weather was so rough that the pilot could not board.
- D. The weather was very rough but the pilot boarded.

2. Choose the word that is most nearly opposite in meaning to "lucid".

- A. Clear
- B. Vague
- C. Logical
- D. Coherent

3. Identify the grammatically correct sentence from the options below.

- A. The captain emphasized the importance of punctuality.
- B. The captain emphasized on the importance of punctuality.
- C. The captain emphasized about the importance of punctuality.
- D. The captain emphasized for the importance of punctuality.

SECTION 5 - NUMERICAL APTITUDE

1. Two workers can complete a job in 10 hours and 15 hours respectively. If they work together, how long will it take them to finish the job?

- A. 6 hours
- B. 9 hours
- C. 12 hours
- D. 7 hours

2. If \$4,000 is invested at 5% per annum simple interest, how much interest will be earned in 4 years?

- A. \$1,000
- B. \$800
- C. \$700
- D. \$600

3. Two boats are anchored on opposite sides of a 100 m tall lighthouse. From the boats, the angles of elevation to the top of the lighthouse are 30° and 45° respectively. What is the distance between the two boats?

- A. 273 m
- B. 173 m
- C. 200 m
- D. 300 m

SECTION 6 - MARITIME KNOWLEDGE

1. When crossing a traffic separation scheme, the vessel should:

- A. Cross at any convenient angle
- B. Cross at right angles to the traffic flow
- C. Alter course to avoid lanes
- D. Follow the traffic flow

2. At what temperature does the air reach saturation, causing water vapor to begin condensing into liquid droplets?

- A. Temperature at which air becomes saturated
- B. Temperature of condensation clouds
- C. Lowest night temperature
- D. Temperature at which rain begins

3. Aneroid barometers are typically calibrated to display atmospheric pressure in which of the following units?

- A. Inches of mercury and millibars
- B. Feet of mercury and millibars
- C. Inches of mercury and millimeters
- D. Inches of mercury and centimeters

4. In the Northern Hemisphere, what is the typical direction of wind circulation around a low-pressure area?

- A. Anticlockwise and inward
- B. Clockwise and inward
- C. Clockwise and outward
- D. Anticlockwise and outward

5. A navigational aid emits light for a shorter time than it remains dark in each cycle. What is this light called?

- A. Isophase
- B. Occulting
- C. Intermittent
- D. Flashing

6. Which factor causes the settling error of a gyrocompass to become more pronounced?

- A. Operating at higher latitudes
- B. Operating at lower latitudes
- C. Increasing the vessel's speed
- D. Navigating in areas of high magnetic variation

7. If 1800 metric tonnes of cargo are loaded, what will be the vessel's new mean draft?

- A. 10.10 m
- B. 9.19 m
- C. 9.80 m
- D. 9.99 m

8. In radar navigation, what is the main purpose of using a parallel index line?

- A. Determining gyro error
- B. Maintaining a constant distance from hazards or the coastline
- C. Plotting tidal set and drift
- D. Calculating the vessel's speed

9. In marine meteorology, what does a rapidly falling barometer usually indicate?

- A. Stable weather conditions
- B. Approaching bad weather or storm
- C. Extremely dry air
- D. Calm sea conditions

10. A vessel departs Felixstowe, England at 1000 hrs local time (GMT+0) on 2nd March 2014, bound for Kolkata, India, a distance of 10,080 nautical miles. If the vessel maintains a constant speed of 10 knots, what will be the vessel's estimated time of arrival (ETA) at Kolkata in local time (GMT+5.5)?

- A. 13th April 2014, 1000 hrs LT
- B. 13th April 2014, 0430 hrs LT
- C. 13th April 2014, 1530 hrs LT
- D. 13th April 2014, 1200 hrs LT

11. The azimuth of a celestial body is measured at which of the following locations?

- A. Observer's Zenith
- B. Observer's Nadir
- C. Observer's Meridian
- D. Observer's Prime Vertical

12. Which instrument is specifically used to measure the apparent wind direction and speed on a moving ship?

- A. Barometer
- B. Psychrometer
- C. Hygrometer
- D. Anemometer

13. Ventilators forming part of Conditions of Assignment must:

- A. Be of any height if fitted with wire mesh
- B. Have minimum height and efficient closing appliances
- C. Have closing devices only above superstructure decks
- D. Be permanently open for machinery ventilation

14. Which cloud type is most commonly associated with thunderstorms and heavy rainfall?

- A. Cirrus
- B. Stratus
- C. Cumulonimbus
- D. Altocumulus

15. The total area beneath the GZ (righting arm) curve from 0° to 40° of heel is a measure of which of the following vessel characteristics?

- A. The vessel's dynamic stability within that heel range
- B. The vessel's metacentric height (GM)
- C. The vessel's KM
- D. The vessel's maximum righting lever

16. When approaching a danger marked by a Cardinal buoy, the vessel should:

- A. Pass on the port side
- B. Pass on the starboard side
- C. Keep to the side named by the buoy
- D. Keep the buoy on the vessel's port side

17. At which location on Earth does the Coriolis effect reach its greatest strength?

- A. At the Equator
- B. At the Tropics
- C. At 30° latitude
- D. At the Poles

18. On which radar display mode does the own ship remain stationary at the center while other vessels and objects appear to move around it?

- A. North Up
- B. Course Up
- C. Relative Motion
- D. True Motion

19. How should a newly issued Preliminary Notice to Mariners (P-Notice) be marked on a nautical chart?

- A. In red marker
- B. Lightly in pencil
- C. In permanent ink
- D. Not marked until confirmed

20. Which celestial body is commonly used for obtaining a compass error by azimuth during twilight?

- A. Polaris
- B. Venus
- C. The Sun
- D. Mars

21. Which gas is primarily responsible for the greenhouse effect?

- A. Oxygen
- B. Nitrogen
- C. Carbon dioxide
- D. Helium

22. Which opening below is NOT considered while assessing Conditions of Assignment?

- A. Hatchway on freeboard deck
- B. Door in superstructure end bulkhead
- C. Sounding pipe opening in engine room
- D. Ventilator on freeboard deck

23. What does SOLAS stand for?

- A. Safety of Life at Sea
- B. Security of Life and Ships
- C. Standard Operations for Large Ships
- D. Safety Operations and Life Saving

24. Which type of weather front is most commonly linked to severe thunderstorms, heavy rain, and sudden gusty winds?

- A. Cold front
- B. Occluded front
- C. Warm front
- D. Stationary front

25. Which frontal type is MOST commonly associated with the process shown in the diagram?

- A. Stationary front
- B. Warm front
- C. Cold front
- D. Occluded front

26. In celestial navigation, what is the First Point of Aries () primarily used to measure?

- A. Measuring True Altitude
- B. Measuring Azimuth
- C. Measuring Right Ascension
- D. Measuring Longitude

27. In celestial navigation, what is the reason for drawing the intercept at right angles to the azimuth direction?

- A. Because the azimuth determines the observer's latitude
- B. Because the Line of Position (LOP) is always perpendicular to the azimuth to the celestial body
- C. Because the intercept is affected by time zones
- D. Because the Earth's magnetic field influences the plotting

28. Which part of the ship is located at the front?

- A. Stern
- B. Bow

- C. Port
- D. Starboard

29. What does the term "angle of loll" describe in ship stability?

- A. The angle at which the center of gravity is vertically above the center of buoyancy
- B. The time taken for a vessel to complete one roll cycle
- C. KM is greater than GM
- D. The heel angle at which a vessel with negative GM achieves equilibrium due to hull shape

30. The observer's prime vertical passes through the east and west points of which of the following?

- A. Rational horizon
- B. Equinoctial
- C. Celestial meridian
- D. Great circle

31. What is the primary purpose of a marine sextant?

- A. Measuring wind direction
- B. Determining celestial altitude
- C. Measuring sea depth
- D. Determining engine power

32. Which part of a ship experiences the greatest bending moment in heavy seas?

- A. Midship section
- B. Forecastle
- C. Stern peak
- D. Bridge deck

33. Which of the following corrections must be applied to a sextant reading to accurately determine the observed altitude (H_o) of a celestial body?

- A. Index error, dip, refraction, semi-diameter & parallax
- B. Index error and dip only
- C. Refraction only
- D. Dip and refraction only

34. On a nautical chart, which color is commonly used to indicate shallow water and the boundaries of certain regulated areas?

- A. Gold
- B. Blue
- C. Black
- D. Green

35. Which instrument is used to measure the salinity of seawater?

- A. Hygrometer
- B. Sextant
- C. Salinometer
- D. Barograph

36. Which navigation light is carried by a vessel not under command?

- A. Two all-round red lights in a vertical line
- B. One all-round white light
- C. Green and red sidelights only
- D. Two all-round green lights

37. In ship stability, GM refers to:

- A. Gross Measurement
- B. General Metric
- C. Metacentric Height
- D. Gravity Measurement

38. Which document contains detailed information about cargo carried on a vessel?

- A. Deck Log Book

- B. Cargo Manifest
- C. Engine Log Book
- D. Stability Booklet

39. What is the primary navigational purpose of observing the Sun at its meridian passage?

- A. Applying time correction
- B. Finding longitude
- C. Calculating compass error
- D. Determining latitude

40. Which safety equipment is used to prevent drowning during emergencies?

- A. Compass
- B. Lifebuoy
- C. Sextant
- D. Radar reflector